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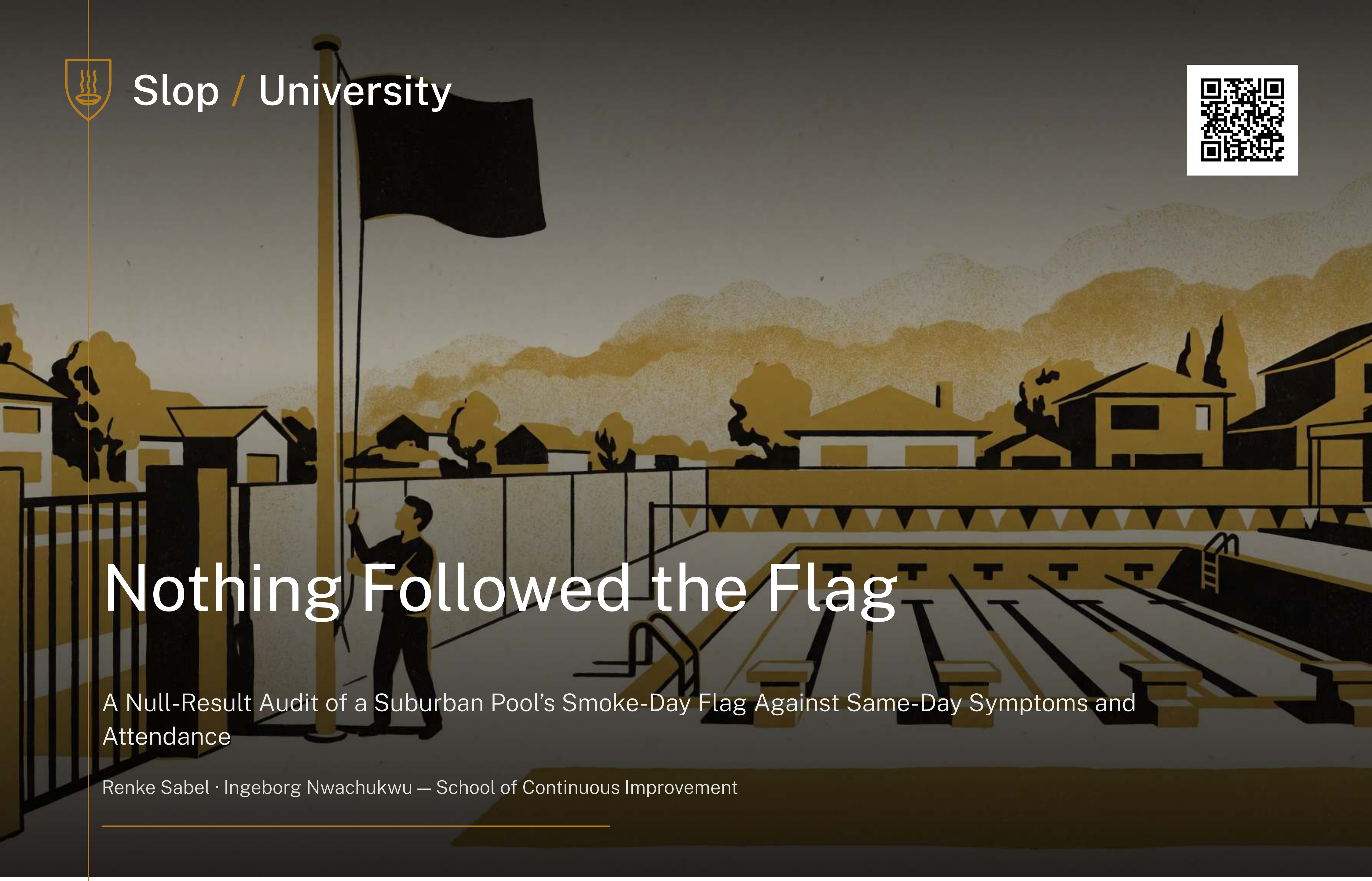

Nothing Followed the Flag

A Null-Result Audit of a Suburban Pool's Smoke-Day Flag Against Same-Day Symptoms and Attendance

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Background

Suburban pools increasingly post a daily smoke-day flag, set each morning by a rostered duty volunteer against a government air-quality index. Whether the flag tracks anything patrons or staff independently notice about that day remains untested.

Aims

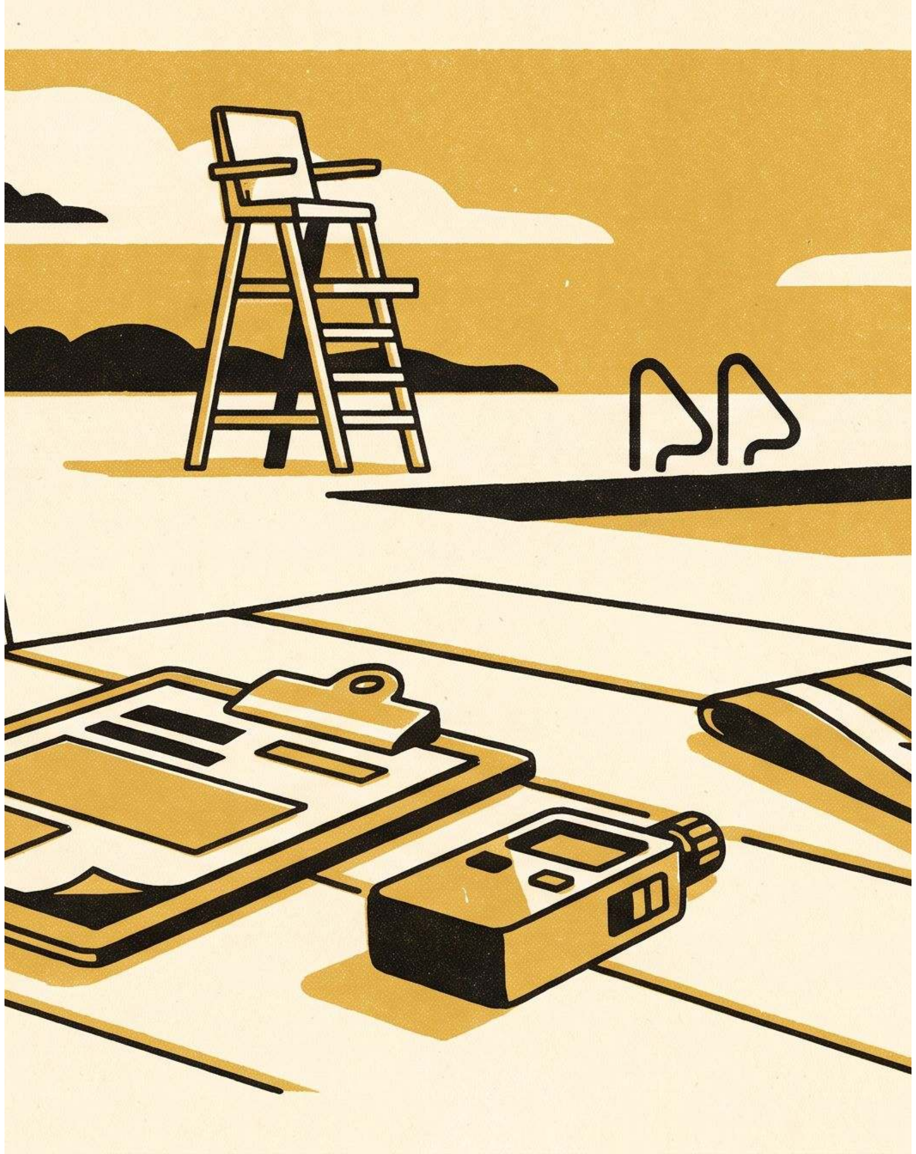
- Test whether flag status predicts same-day patron-reported symptoms
- Test whether flag status predicts gate-logged attendance
- Test whether the underlying air-quality reading outperforms the flag itself

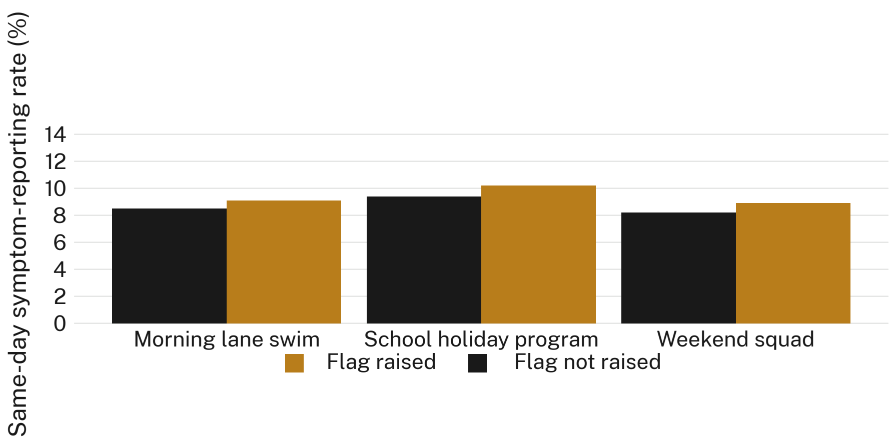
Methods

- 402 duty mornings · 6 suburban pools · 9-week smoke-affected season
- Same-day exit-gate symptom survey, n = 2,183 patrons, collected blind to flag status
- Continuous air-quality reading re-scored against the same mornings; thresholds pre-registered

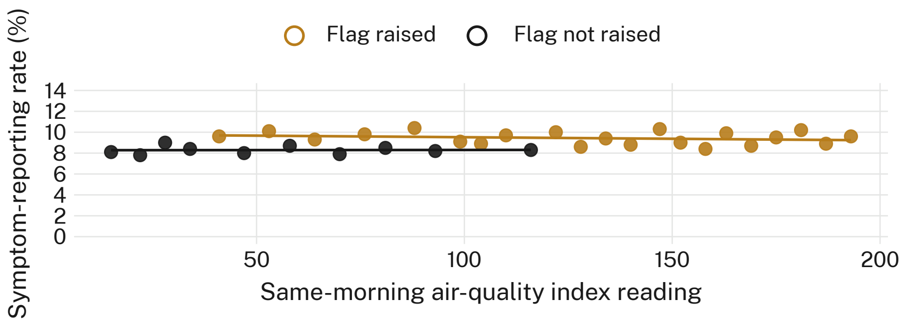
Results

Symptom-reporting holds flat between flag-up (9.4%) and flag-down (8.7%) mornings. Gate-tallied attendance and a continuous re-scoring of the underlying index show no stronger association.





Same-day symptom-reporting rate held within 1.3 percentage points of the flag-down baseline across all three programs, 4 March – 3 May 2026 (n = 402 duty mornings).



Flag-raised and flag-not-raised mornings fit near-identical flat lines against the continuous index reading; symptom-reporting rate tracks neither the flag nor the reading it is drawn from.

Discussion & conclusions

No measure tested — reported symptoms, attendance, the index reading itself — moved with the flag. The flag may still change what a morning contains without tracking what a morning contains. A pre-registered subgroup pass (weekday vs weekend, morning vs afternoon sessions) turns up nothing further. Next: audit the flag's effect on enrolment and refund requests, not its correlation with exposure.

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We thank the duty volunteers of the six participating pools and the coordinating aquatics trust. Symptom surveys de-identified at collection; flag and index logs merged only by date. Thresholds pre-registered; none adjusted post hoc.